

assignment nomentum I

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 8:02pm

Quiz Instructions

tutorial in folder



Question 1 7 pts

A 0.14-kg baseball is dropped from rest from a height of 2.0 m above the ground. What is the magnitude of its momentum just before it hits the ground if we neglect air resistance?

hint: use kinematics to find the speed before it hits the floor. Remember

$v_f^2 = v_i^2 - 2(9.8)h$. then use the definition of momentum

☐

0.28 kg · m/s

☐

0.44 kg · m/s

☐

0.88 kg · m/s

☐

1.4 kg · m/s

☐

0.62 kg · m/s



Question 2 7 pts

A 0.140-kg baseball is dropped and reaches a speed of 1.20 m/s just before it hits the ground and bounces. It rebounds with an upward velocity of 1.00 m/s. What is the change of the ball's momentum during the bounce?

hint: remember the sign. up is + and down is -.

☐

0.0280 kg · m/s upwards

☐

0.000 kg · m/s

☐

0.308 kg · m/s upwards

☐

0.308 kg · m/s downwards



0.0280 kg · m/s downwards



Question 3 7 pts

A 1200-kg cannon suddenly fires a 100-kg cannonball at 35 m/s. What is the recoil speed of the cannon? Assume that frictional forces are negligible and the cannon is fired horizontally.



2.9 m/s



3.2 m/s



3.5 m/s



35 m/s



Question 4 7 pts

A 1200-kg car moving at 15.6 m/s suddenly collides with a stationary car of mass 1500 kg. If the two vehicles lock together, what is their combined velocity immediately after the collision?



6.9 m/s



5.5 m/s



12.1 m/s



8.6 m/s



Question 5 7 pts

A girl of mass 55 kg throws a ball of mass 0.80 kg against a wall. The ball strikes the wall horizontally with a speed of 25 m/s, and it bounces back with this same speed. The ball is in contact with the wall 0.050 s. What is the magnitude of the average force exerted on the wall by the ball?



13,750 N



400 N



800 N



27,500 N



55,000 N



Question 6 7 pts

A 1000-kg whale swims horizontally to the right at a speed of 6.0 m/s. It suddenly collides directly with a stationary seal of mass 200 kg. The seal grabs onto the whale and holds fast. What is the momentum of these two sea creatures just after their collision? You can neglect any drag effects of the water during the collision.

hint: the total momentum before = total momentum after

☐

7200 kg · m/s

☐

1200 kg · m/s

☐

2000 kg · m/s

☐

0.00 kg · m/s

☐

6000 kg · m/s

☐

Question 7 7 pts

A 1000-kg whale swims horizontally to the right at a speed of 6.0 m/s. It suddenly collides directly with a stationary seal of mass 200 kg. The seal grabs onto the whale and holds fast. What is the speed of these two sea creatures just after their collision? You can neglect any drag effects of the water during the collision.

☐

5.0 m/s

☐

4.0 m/s

☐

3.0 m/s

☐

6.0 m/s

☐

0.00 m/s

☐

Question 8 7 pts

A 480-kg car moving at 14.4 m/s hits from behind a 570-kg car moving at 13.3 m/s in the same direction. If the new speed of the heavier car is 14.0 m/s, what is the speed of the lighter car after the collision, assuming that any unbalanced forces on the system are negligibly small?

hint: use $m_1V_1 + m_2V_2 = m_1V_1' + m_2V_2'$

☐

5.24 m/s

☐

19.9 m/s

☐

13.6 m/s

☐

10.5 m/s

**Question 9 7 pts**

Astronaut Jennifer's lifeline to her spaceship comes loose and she finds herself stranded, "floating" 100 m from the mothership. She suddenly throws her 2.00-kg wrench at 20 m/s in a direction away from the ship. If she and her spacesuit have a combined mass of 200 kg, how long does it take her to coast back to her spaceship?

hint: Its a recoil situation. First find Jen's speed (like the ice skaters). Then use kinematics $\text{speed} = \text{distance}/\text{time}$ to find the time. The speed is constant. No friction.

☐

750 s

☐

2.50 min

☐

1000 s

☐

250 s

☐

500 s

**Question 10 7 pts**

Two ice skaters suddenly push off against one another starting from a stationary position. The 45-kg skater acquires a speed of 0.375 m/s relative to the ice. What speed does the 60-kg skater acquire relative to the ice?

☐

0.75 m/s

☐

0.50 m/s

☐

0.38 m/s

☐

0.28 m/s

☐

0.00 m/s

**Question 11 7 pts**

A small car meshes with a large truck in a head-on collision. Which of the following statements concerning the magnitude of the momentum change during the collision is correct? (There could be more than one correct choice.)

☐

The small car and the truck experience the same magnitude momentum change.

☐

The truck experiences the greater magnitude momentum change.

☐

The small car experiences the greater magnitude momentum change.

☐

The magnitude of the momentum change experienced by each one is directly proportional to its mass.

☐

The magnitude of the momentum change experienced by each one is inversely proportional to its mass.



Question 12 7 pts

Which of the following quantities are units of momentum? (There could be more than one correct choice.)

☐

$\text{kg} \cdot \text{m/s}$

☐

$\text{N} \cdot \text{s}$

☐

$\text{N} \cdot \text{m}$

☐

$\text{kg} \cdot \text{m}^2/\text{s}^2$

☐

$\text{kg} \cdot \text{s/m}$



Question 13 7 pts

Two ice skaters push off against one another starting from a stationary position. The 45.0-kg skater acquires a speed of 0.375 m/s. What speed does the 60.0-kg skater acquire? Assume that any other unbalanced forces during the collision are negligible.

☐

0.281 m/s

☐

0.000 m/s

☐

0.375 m/s

☐

0.750 m/s

☐

0.500 m/s



Question 14 7 pts

A rubber ball and a lump of clay have equal mass. They are thrown with equal speed against a wall. The ball bounces back with nearly the same speed with which it hit. The clay sticks to the wall. Which one of these objects experiences the greater momentum change?

☐

the clay

☐

Both of them experience zero momentum change.

☐

the ball

☐

Both of them experience the same non-zero momentum change.

☐

Question 15 7 pts

go

https://www.walter-fendt.de/html5/phen/collision_en.htm  (https://www.walter-fendt.de/html5/phen/collision_en.htm)

Wagon 1:		
Mass:	1.0	kg
Velocity:	0.2	m/s
Wagon 2:		
Mass:	1.0	kg
Velocity:	-0.2	m/s

The 2 cars collide with each other. Whats happening

☐

They exchange their velocity

☐

They both stop after collision

☐

One stop and the second one has twice the velocity

☐

One slows down and the second speeds up

☐

Question 16 1 pts

Building on the previous question:

Wagon 1:		
Mass:	0.1	kg
Velocity:	0.2	m/s
Wagon 2:		
Mass:	1.0	kg
Velocity:	0.0	m/s

☐

they collide and move both in the same direction

☐

both cars bounce each other

☐

the red car bounces off and the blue car moves slowly @right

☐

the red car stops and the blue car moves

Not saved

Submit Quiz